RV College of Engineering®

Unit-I
FUELS

Go, change the world®

Outline

- Energy is necessary to do work and is produced when fuel undergoes combustion.
- Study deals with different types of calorific values, knocking in petrol and diesel engine, knocking in terms of octane no and cetane no and solutions to overcome adverse effect of knocking.
- As it is known after few years coal and petroleum will not be available, so we will be discussing some inexhaustible source of energy.
- A special category of fuel and oxidizer is required to undergo combustion to give high thrust to rocket. Briefly we will discuss about the rocket propellant types, characteristics and properties.

Thermo Chemistry

- Thermochemistry deals with the study of heats of chemical reactions.
- During chemical reaction, energy in various forms may be emitted or absorbed.
- During many reactions the temperature rises indicating that heat is being evolved. In others, when temperature falls, absorption of heat is indicated.
- Thus, the study of the heat produced or required by chemical reaction forms the basis of thermochemistry.
- The reactions in which heat escapes from the system to the surroundings are termed as exothermic process.

Thermo Chemistry

- The chemical processes in which heat is absorbed by the system from the surroundings are endothermic process.
- The absorption of heat leads to increase in the enthalpy of a system at constant pressure, thus the enthalpy change is positive.

Fuel

- A fuel is defined as naturally occurring or artificially manufactured combustible carbonaceous material which serves particularly as source of heat and light.

$$\text{Fuel} + O_2 \xrightarrow{\text{high temp}} \text{Combustion products} + \text{Heat}$$

- As most of the fuels contain carbon or carbon and hydrogen, the combustion involves the oxidation of carbon-to-carbon dioxide and hydrogen to water.

- Sulphur, if present, is oxidized to sulphur dioxide.

- The mineral matter forms the ash.

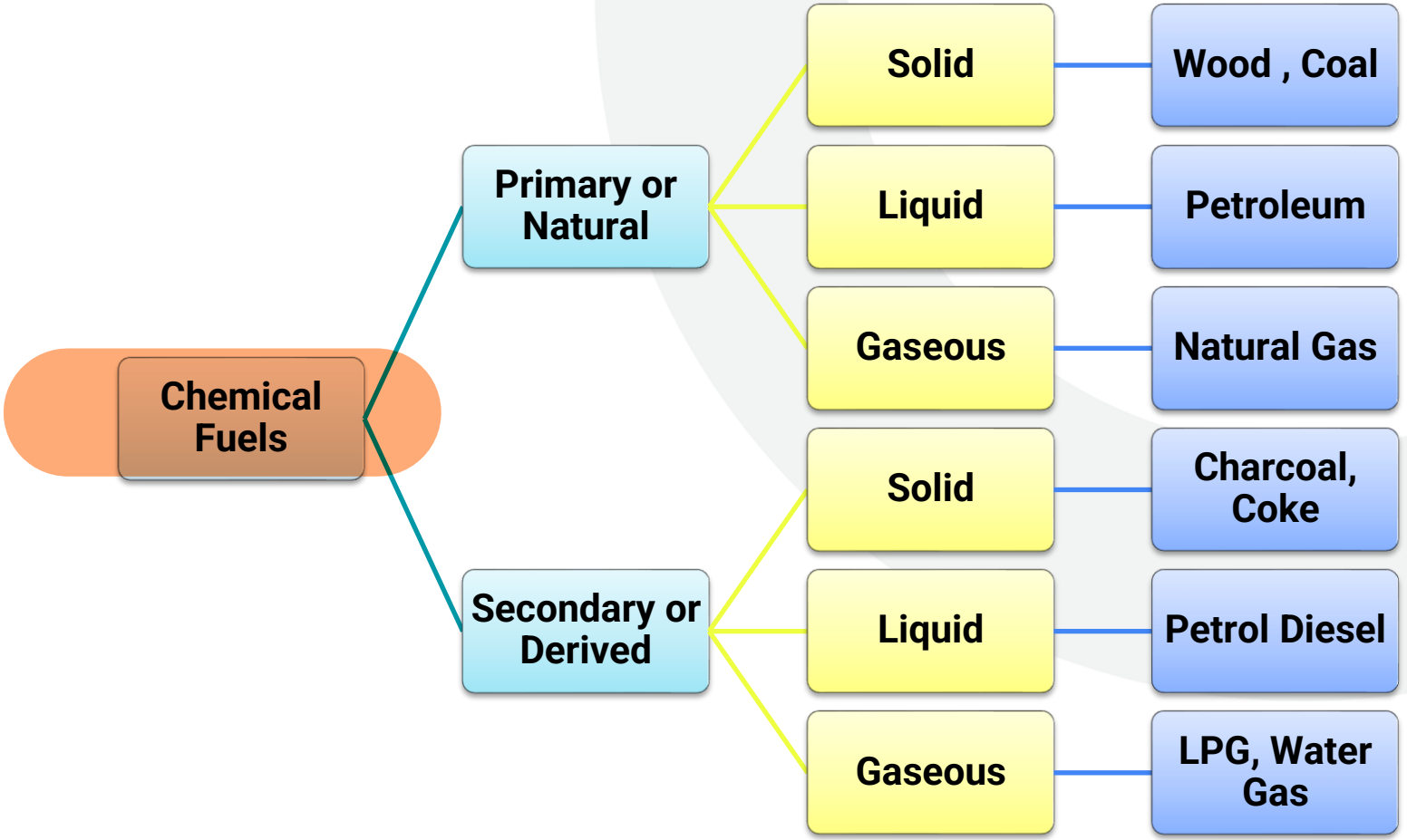
Importance of hydrocarbons as fuels

- Fossil fuels contain mainly hydrocarbons.
- Important sources of energy in daily life.
- Hydrocarbons are used as energy sources in cooking, lighting, automobiles, production of electricity in thermal power plants etc.
- These hydrocarbon fuels meet 80% of the world's energy demand. Thus, hydrocarbons are important sources of energy.

Ideal fuels

- It should possess high calorific value.
- It should have proper ignition temperature. The ignition temperature of the fuel should neither be too low nor too high.
- It should not produce poisonous products during combustion.
- It should have moderate rate of combustion.
- Combustion should be easily controllable i.e., combustion of fuel should be easy to start or stop as and when required.
- It should not leave behind much ash on combustion.
- It should have low moisture content and should be cheap.

Classification of fuels



Calorific value of fuels

- Calorific value is defined as the amount of heat liberated when a unit mass of fuel is burnt completely in presence of air or oxygen.

- Types:

Higher calorific value. (HCV) or Gross calorific value. (GCV)

Lower calorific value. (LCV) or Net calorific value. (NCV)

Calorific value of fuels

- **HCV / GCV:** It is the amount of heat liberated when a unit mass of fuel burnt completely in the presence of air or oxygen and the products of combustion are cooled to room temperature. Here it includes the heat liberated during combustion and the latent heat of steam.

- **LCV / NCV:** It is amount of heat liberated when a unit mass of fuel is burnt completely in the presence of air or oxygen and the product of combustion are let off completely into air. It does not include the latent heat of steam. Therefore, it is always lesser than HCV.

Gross calorific value = Net Calorific value + Heat associated with steam

Calorific value of fuels

Calculation of Heat associated with steam

Let the percentage of hydrogen in the fuel be “H”

2 kg of hydrogen produces 18 kg of water

Therefore, H kg of Hydrogen produces (18/2) X H kg of water

Water produced by 1 kg of fuel = (18/2) x (H/100) kg = 0.09 H kg

Heat associated with steam = weight of water produced x Latent heat of condensation or vapourization

= 0.09H x 587 x 4.2 kJ/kg

(587 is the latent heat of condensation in cal /gram)

Gross calorific value = Net Calorific value + Heat associated with steam

Calorific value of fuels

Find out the percentage of Hydrogen in the 5 g of Hydrocarbon fuel burnt in air and heat carried away by H_2O is 250 kJ/kg

Ans: Heat associated with steam = 0.09H x 587 x 4.2 kJ/kg

H = 1.126

Calorific value of fuels

3.1 kg o fuel was burnt in air and combustion products were allowed to cool at 298K. The calorific value obtained by the process is 500 kJ/kg. If the % of hydrogen in the fuel is 2 then what is the net calorific value.

Ans: % of hydrogen = 2

100 kg fuel contains 2 kg of hydrogen

GCV = 500 kJ/kg

= 56.22 kJ/kg

Calorific value of fuels

5 kg of fuel contains 0.25 kg of hydrogen. Same fuel burnt in presence of air completely and produced NCV of 500 kJ/kg. Calculate GCV

Ans: 5 kg fuel contains 0.25 kg of hydrogen

H = (0.25*100)/5 = 5 kg

Heat associated with steam = 0.09H x 587 x 4.2 kJ/kg

=0.09*5*587*4.2 kJ/kg = 1109.43 kJ/kg

GCV = (500 +1109.43) kJ/kg = 1609.43 kJ/kg

Calorific value of fuels

5 kg of fuel contains Carbon, hydrogen and Oxygen (5:1:2). Same fuel burnt in presence of air completely and produced NCV of 500 kJ/kg. Calculate GCV.

Ans: (5:1:2)X = 100, 8X = 100, X = 12.5

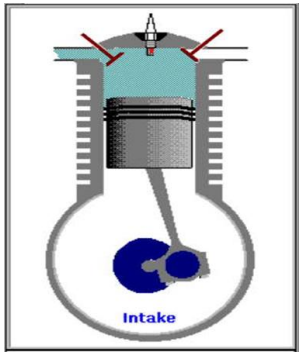
C % = 62.5, H % = 12.5, O % = 25

Heat associated with steam = 0.09H x 587 x 4.2 kJ/kg

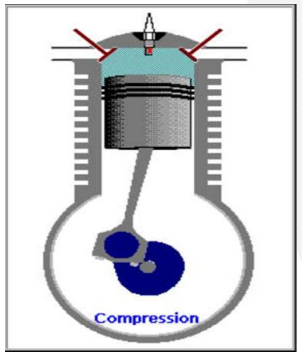
=0.09*12.5*587*4.2 kJ/kg = 2773.575 kJ/kg

GCV = (500 +2773.575) kJ/kg = 3273.575 kJ/kg

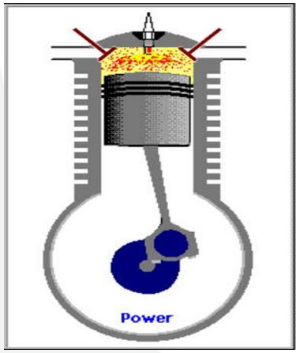
Knocking in Petrol Engine



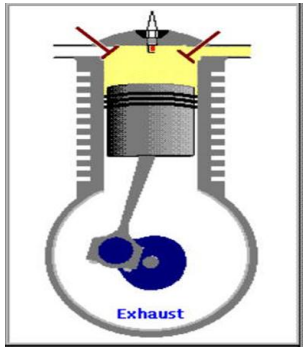
Suction Stroke



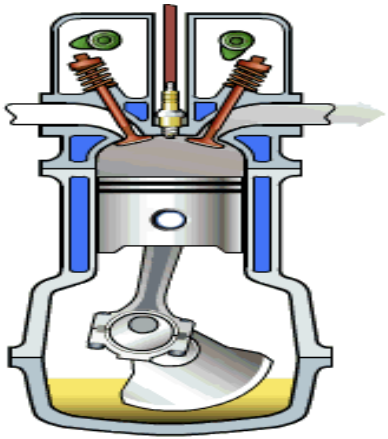
Compression Stroke

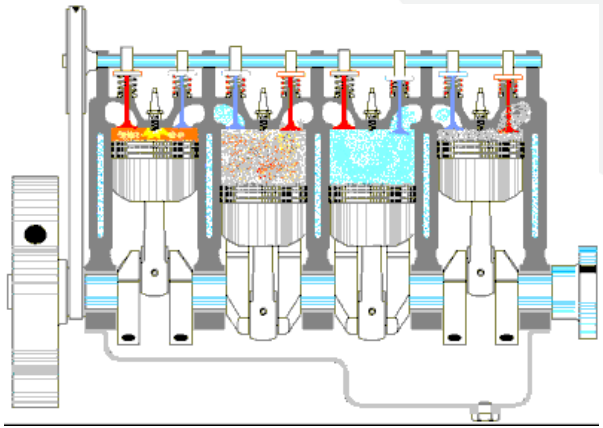


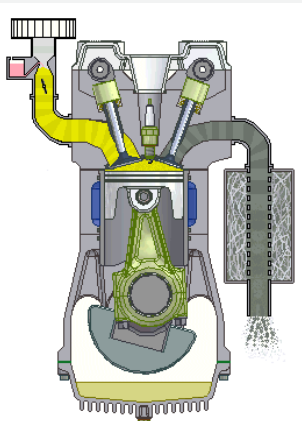
Power Stroke



Exaust Stroke

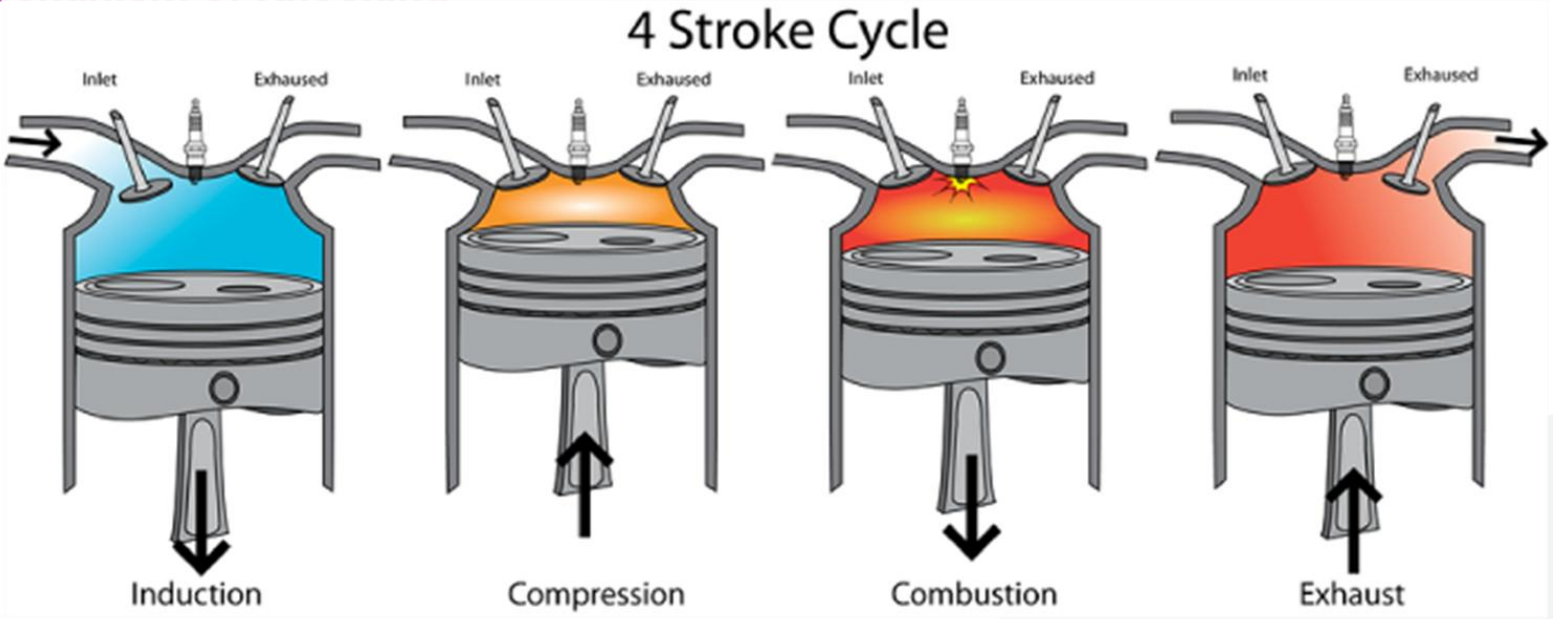






Knocking in Petrol Engine

Mechanism of Knocking



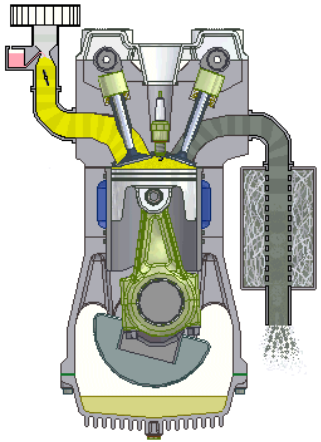
"Knocking is the rattling noise or metallic sound produced in an internal combustion engine due to uneven combustion of the fuel and air mixture"

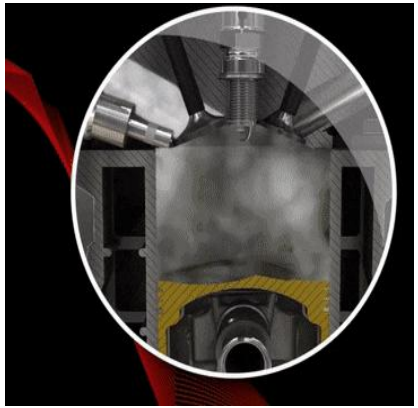
Reasons for Knocking

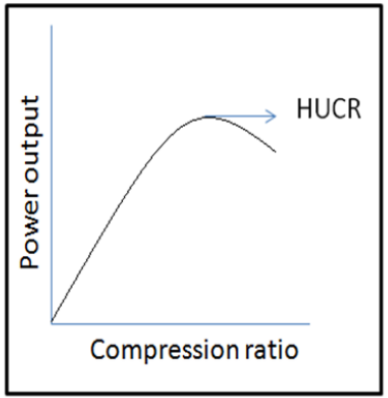
Very High Compression Ratio- Pre ignition

Compression ratio is the ratio of the volume of the fuel and air mixture (V1) at the end of suction stroke to the volume of the mixture (V2) at the end of compression stroke.

Compression ratio (CR) = V1/V2







- Hot Spots
- Highest Useful Compression ratio (HUCR)
- Pre-ignition
- Knocking

Reasons for Knocking

Peroxide formation leads to self Ignition

Under normal conditions there is a slow oxidation of the fuel. Normal combustion takes place due to the chain reaction. The overall reaction is,

$$C_2H_6 + 3/2O_2 \longrightarrow 2CO_2 + 3H_2O$$

Under knocking conditions, the rate of combustion is very high.

1. Oxygen combines with hydrocarbon molecule forming peroxides.

$$H_3C{-}CH_3 + O_2 \longrightarrow H_3C{-}O{-}O{-}CH_3$$

Ethane — Ethane peroxide

2. The peroxides decompose readily to give a number of gaseous products. For example,

$$H_3C{-}O{-}O{-}CH_3 \longrightarrow CH_3{-}CHO + H_2O$$

Ethane peroxide — Acetaldehyde

$$CH_3{-}CHO + 1\,1/2O_2 \longrightarrow HCHO + CO_2 + H_2O$$

Acetaldehyde — Formaldehyde

$$HCHO + O_2 \longrightarrow H_2O + CO_2$$

Formaldehyde

Reasons for Knocking

Chemical structure of hydrocarbon

Knocking tendency decreases with the increase in compactness of the molecule and increases with the increase in the length of hydrocarbon chain.

i) 2, 2-dimethyl pentane knocks lesser than 2-methyl hexane

ii) n-butane knocks lesser than n-hexane.

The tendency to knock has been found to decrease in the order

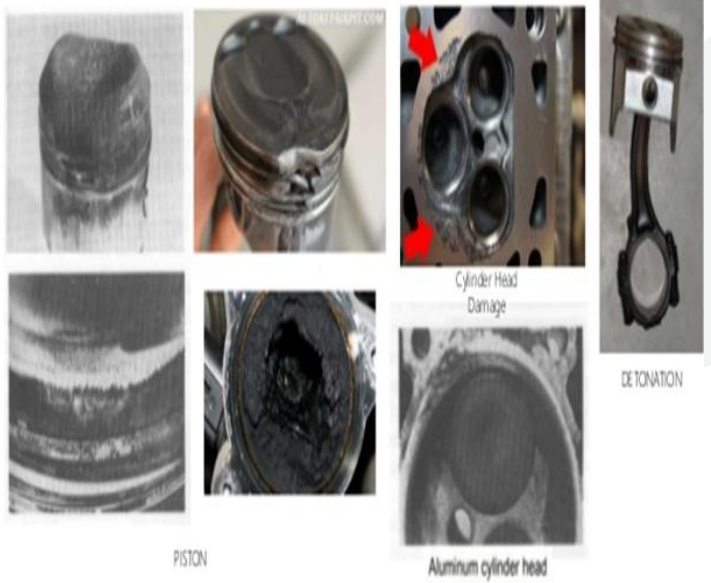
n-alkane > mono substituted alkanes > cycloalkane > alkenes> Polysubstituted alkanes >aromatic compounds.

Effects of Knocking

- Noise and Roughness
- Vibration
- Mechanical damage
- Decrease in power
- Pre-ignition



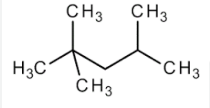




Octane Number

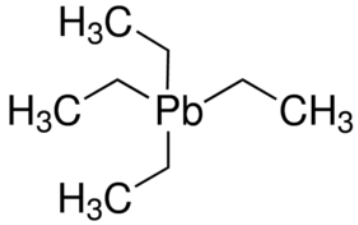
Graham Edger proposed an arbitrary scale for expressing the knocking characteristics of gasoline in combustion engine as octane number in 1972 on the basis of antiknock properties.

"Octane number of a petrol or gasoline is the percentage of isooctane in a mixture of n-heptane and iso-octane which has same knocking characteristics as that of petrol".

Reference fuels Chemical structure	Octane number
2,2,4 –trimethylpentane  Isooctane	100
CH3-CH2-CH2-CH2-CH2-CH_2-CH_3 n-heptane	0

Anti-Knocking agents

An extensively used anti-knocking agent is tetraethyl lead $Pb(C_2H_5)_4$



Dis-advantages:

1. Lead is poisonous.
2. Deposits on the walls of the cylinder

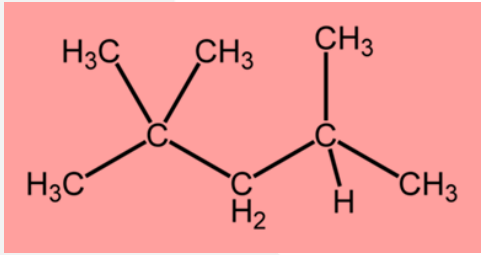
Remedial measures:

Addition of ethylene dibromide, ethylene dichloride, along with TEL

Residual Pb and PbO are converted in to volatile $PbCl_2$ and $PbBr_2$, which will escape with exhaust gases

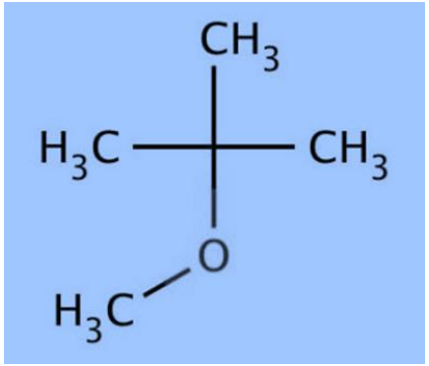
Unleaded petrol

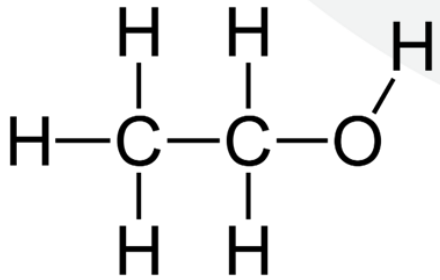
Addition of alkylate fraction to petrol i.e. branched chain alkane of higher mol. wt.



Addition of benzene, toluene, xylene (BTX) to petrol.

Addition of methyl t-butyl ether (MTBE) or ethyl t-butyl ether.





Power Alcohol

Power alcohol is a liquid mixture containing 80% petrol, 20% **ethanol**.

It is useful in the field of automobiles to generate **energy** for its operations

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{invertase}} 2C_6H_{12}O_6$$

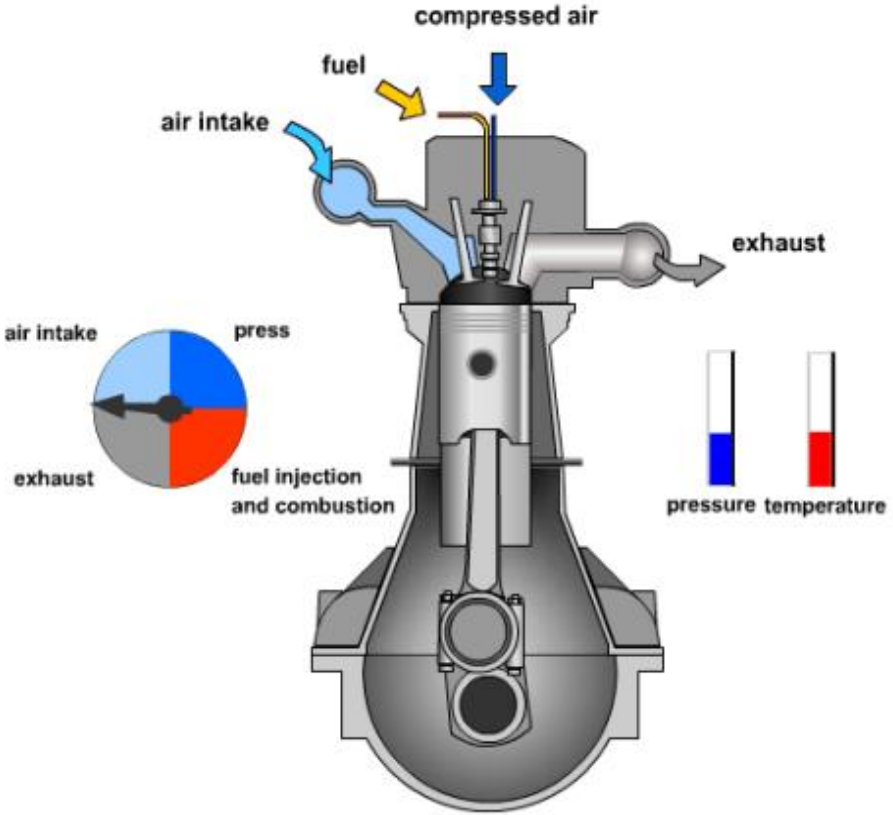
Advantages of Power Alcohol

- Ethyl alcohol has good antiknocking property and its octane number is 90, while the octane number of petrol is about 65. Therefore, addition of ethyl alcohol to petrol increases its octane number.
- Alcohol has property of absorbing any traces of water if present in petrol.
- Ethyl alcohol contains 'O' atoms, which help for complete combustion of power alcohol and the polluting emissions of CO, hydrocarbon, particulates are reduced largely.
- Use of ethyl alcohol in petrol reduces our dependence on foreign countries for petrol and saves foreign currency considerably.
- Power alcohol is cheaper than petrol.

Disadvantages of Power Alcohol

- Ethyl alcohol may undergo oxidation to form acetic acid, which corrodes engine parts.
- Ethyl alcohol obtained by fermentation process directly cannot be mixed with petrol, but it must be dehydrated first.
- As ethyl alcohol contains ‘O’ atoms, the amount of air required for complete combustion of power alcohol is lesser and therefore carburettor and engine needs to be adjusted or modified, when only ethyl alcohol is used as fuel.

Working of Diesel Engine-Ignition delay



Suck the air

Compress it

Inject the fuel

Fuel gets ignited

Time gap
Ignition delay

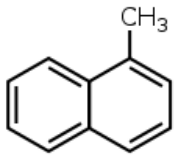
Shorter Ignition delay-
Smooth Combustion

Longer Ignition delay-
Explosive combustion-
Knocking

Cetane Number

Cetane number is a measurement of the combustion quality of diesel fuel during compression ignition.

"Cetane number (or rating) of a diesel is the percentage of cetane in a mixture of cetane and alfa methyl naphthalene, which matches the fuel under test in ignition characteristics"

Reference fuels	Chemical structure	Cetane number
Cetane(hexadecane)	CH_3-$(CH_2)_{14}$-CH_3	**100**
α-methyl naphthalene		**0**

Relative order of cetane no. values

alkanes > cycloalkanes > alkenes > branched alkanes > Aromatics

How to increase Cetane No.

Alkyl nitrates (principally 2-ethylhexyl nitrate) and di-tert-butyl peroxide are used as additives to raise the cetane no.

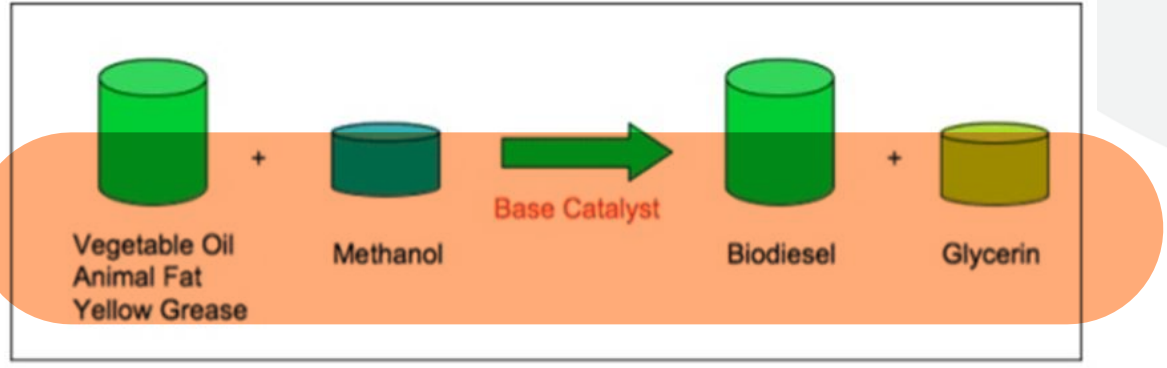
- Antioxidant to improve oxidation resistance during storage.
- Lubricity additives for lubrication of fuel injection system.
- Detergents to clean fuel line and carburetor.

Biodiesel


Who says money don't grow on trees?
BIO DIESEL
DIESEL
Thomas Yoon, Butterworth
http://www.cartoon-web.com

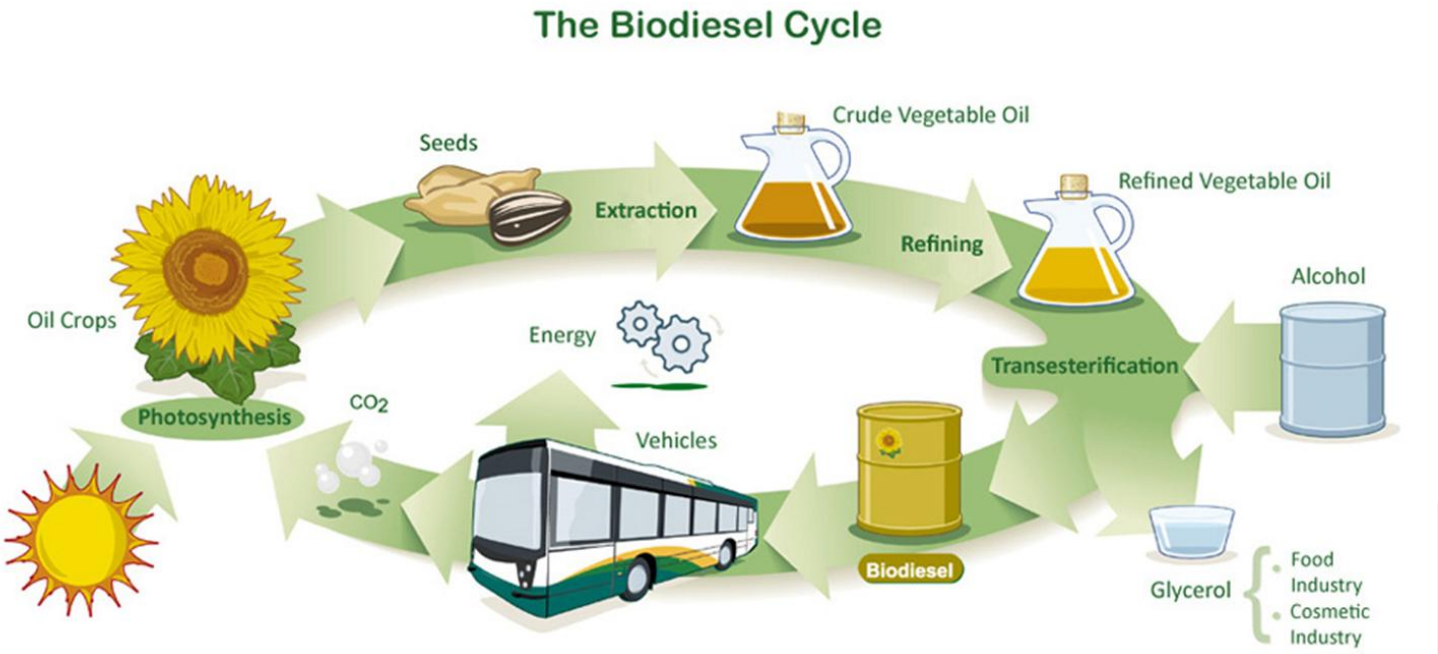





Vegetable Oil
Animal Fat
Yellow Grease
+
Methanol
Base Catalyst
Biodiesel
+
Glycerin



Biodiesel



- Natural, renewable sources such as **vegetable oil** and fats.

- The most commonly used oils for the **production of Biodiesel** are **soybean**, sunflower, palm, cotton seed, **Jatropha, Pongamia.**

Biodiesel





Jatropha

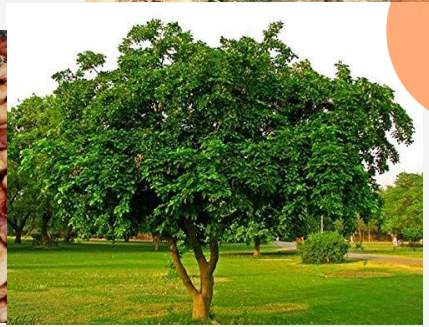




Palm









Ground nut



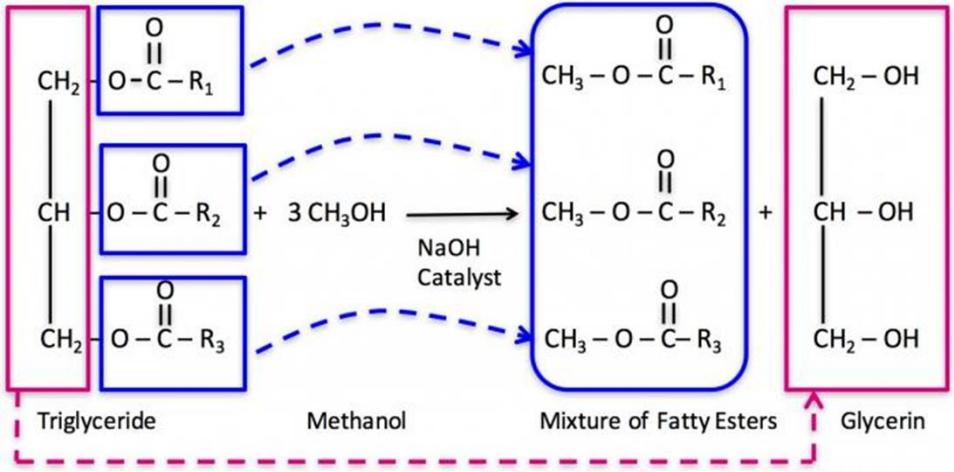
Soyabean



Sun flower

Biodiesel

- Alternative fuel for diesel engines.
- Made from vegetable oil or animal fat .
- Biodegradable, Essentially non-toxic.
- Chemically, biodiesel molecules are mono-alkyl esters produced usually from triglyceride esters



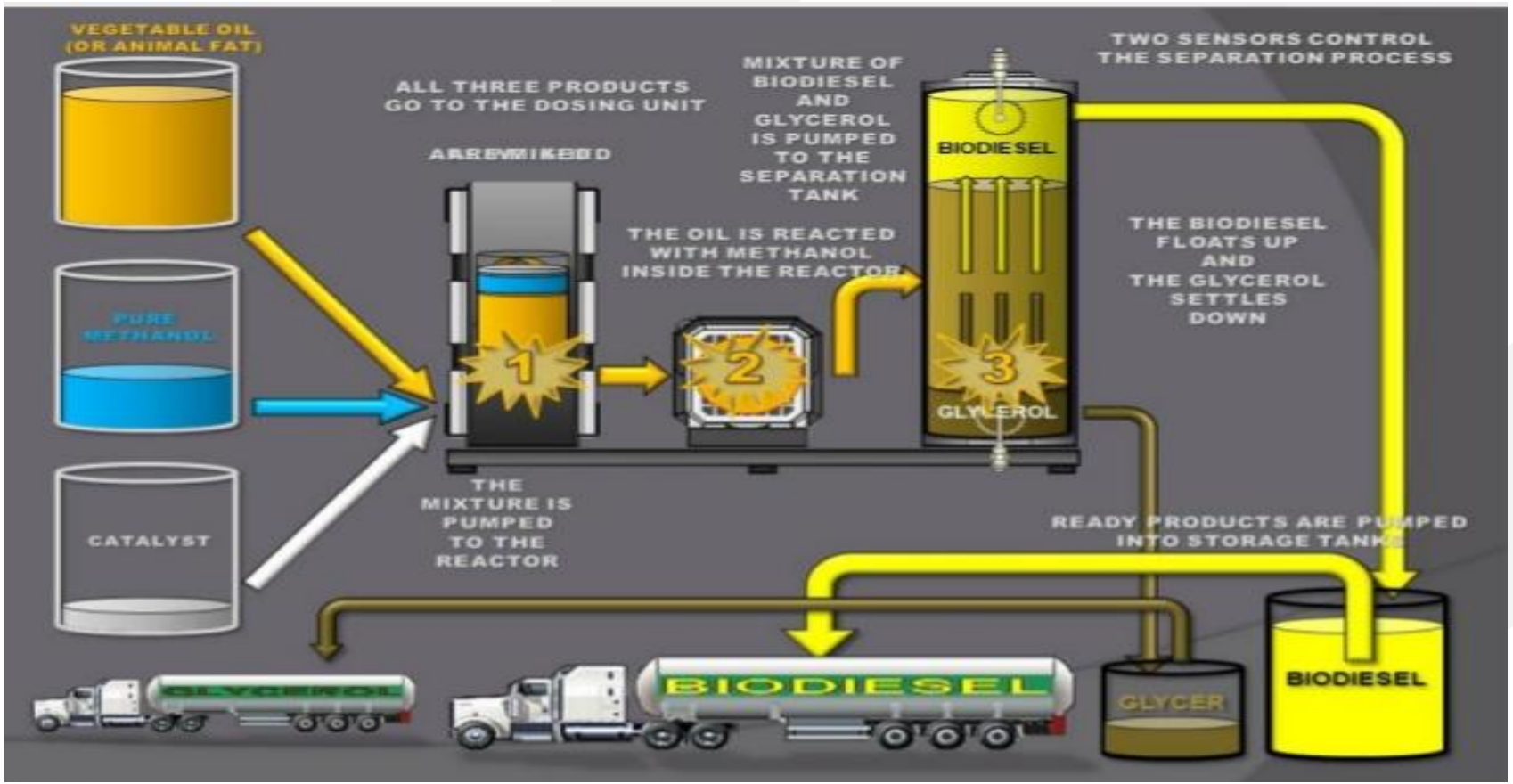
Trans esterification

Reaction

Biproduct

Washing with water

Biodiesel Synthesis



Advantages of Biodiesel

1. Biodiesel is cheaper, as it is manufactured from cheap, non-edible or waste oil or animal fats.
2. It has high cetane numbers 46 to 54 and high C.V of about 40 kJ/kg.
3. It is environment friendly.
4. It does not give out particulate and CO pollutants, as Oxygen atoms in biodiesel help for complete combustion.
5. Reduces our dependence on diesel on foreign countries, saving currency.

Dis-advantages of Biodiesel

It cannot be used in cold regions.

Biodiesel may have dissolving action rubber hoses, gaskets.

There is shortage of vegetable oils and the starting material if costly, the biodiesel will be costly.

Biodiesel strongly adheres on metals and can become gummy.

Green Fuels

- Green fuels are the fuels which are actively produced from renewable sources and mainly make use of elements such as green hydrocarbons.
- These fuels are produced with the help of several processes that include both thermo chemical as well as biological processes and mainly make use of biomass as their source.
- Types of Green fuels include, Biomass, Geothermal energy, Wind power, Solar energy, Hydro energy, Hydrogen etc.
- Amongst all green fuels hydrogen is the best alternative. Hydrogen is a zero-emissions source of fuel for trains, buses, and cars.
- Hydrogen can be used as a feedstock gas for industries such as chemical, refining and steel. In addition, it is a source of heat and power for buildings and can buffer energy generated from renewable sources.

Hydrogen energy

Why hydrogen ?

- Clean – no greenhouse gases
- Energy security – can be produced from abundant sources
- Economic growth
- Efficient – fuel cells ~75% efficiency
- Portable: Car tanks, micro fuel cells…



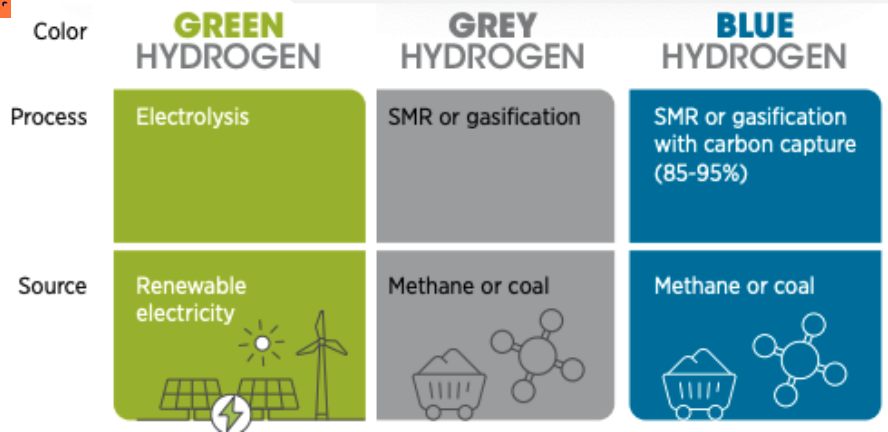
Honda FCX Clarity

Types of Hydrogen

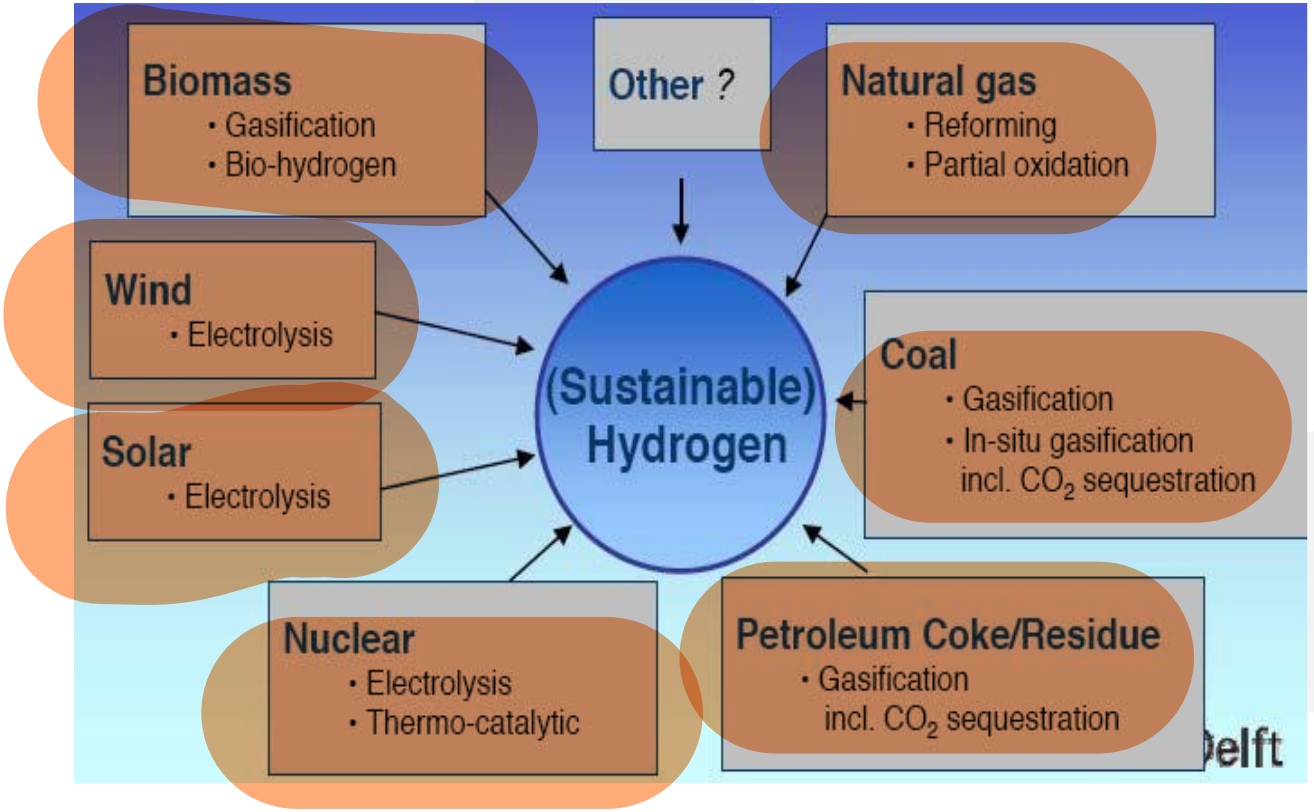
- **Green Hydrogen:** It is produced using electrolysis of water with electricity generated by renewable energy.

- **Grey hydrogen** is created from natural gas, or methane, using steam reformation but without capturing the greenhouse gases made in the process. Most environmental damaging.

- **Blue Hydrogen:** is also produced by methane/natural gas, but the carbon generated from steam reforming is captured and stored underground through industrial carbon capture and storage (CSS).



Hydrogen production



Production of Hydrogen

- Photocatalytic water splitting: Green Hydrogen
- Water splitting is the process in which water will split into hydrogen and oxygen.
- Why to split Water??? Water is most abundant and 70% of the earth is surrounded by water.
- Why photocatalytic process??? Green method. No formation of green house gases. Thermal splitting of water occurs at around 200C.
- Requirements for photocatalyst:

The catalyst having band gap more than 1.23 ev and sufficiently small to make efficient use of solar spectrum (<3eV), Highly crystalline, and highly stable.

What is Photocatalytic water splitting?

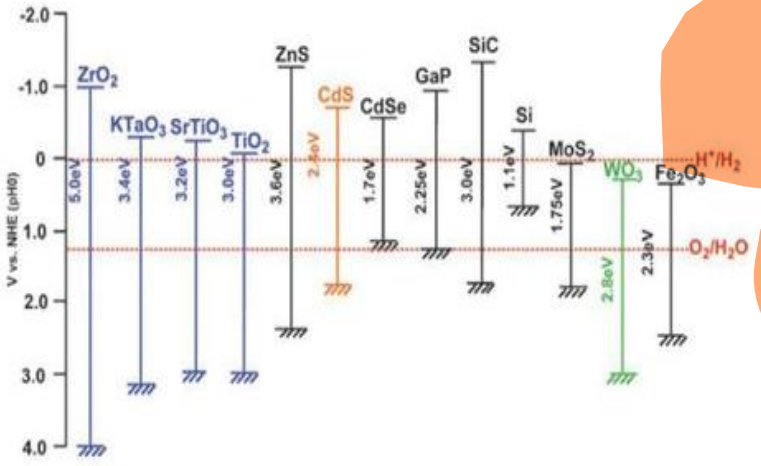
- Water splitting: $H_2O \xrightarrow{h\nu} H_2 + \frac{1}{2}O_2$
  - H_2: clean fuel——no pollution or green-house effect gas.
  - "O_2": water/air purification——degrade and mineralize organics into CO_2 and harmless inorganic anions.
  - The principle may be used to make primary battery.
- Photo: convert solar energy into chemical energy
  - Renewable
  - Environment friendly
  - Economic

Photo catalysts

For electrochemical decomposition of water, a potential difference more than 1.23V is needed.

This is equivalent to a photon of $\lambda \sim 1000$ nm (infrared light).

Requirements for photocatalysts



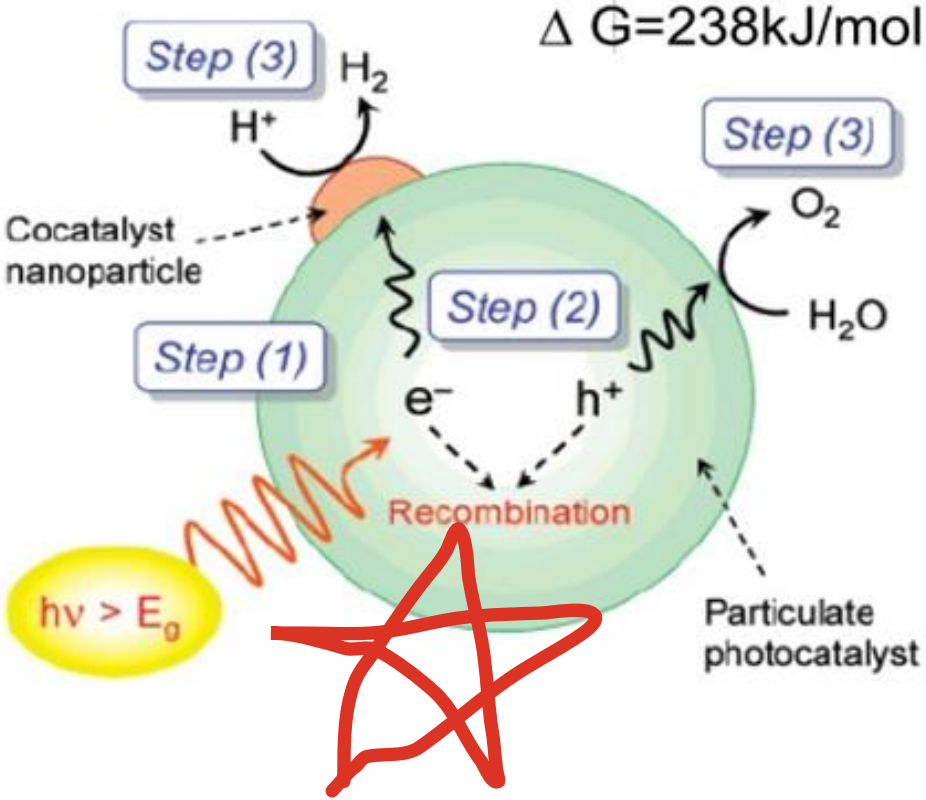
Difficulty to overcome

- Recombination of $e^- - h^+$ pair
- Fast drawback reaction
- Inability to utilize visible light
- Catalyst decay

Mechanism of water splitting

$H_2O \rightarrow 2H_2 + O_2$ $\Delta V = 1.23V$, $\Delta G = 238 kJ/mol$

- Step 1: Photon with energy above 1.23eV (λ<~1000 nm) is absorbed.
- Step 2: Photoexcited electrons and holes separate and migrate to surface.
- Step 3: Adsorbed species (water) is reduced and oxidized by the electrons and holes.



Photocatalytic water splitting: Mechanism

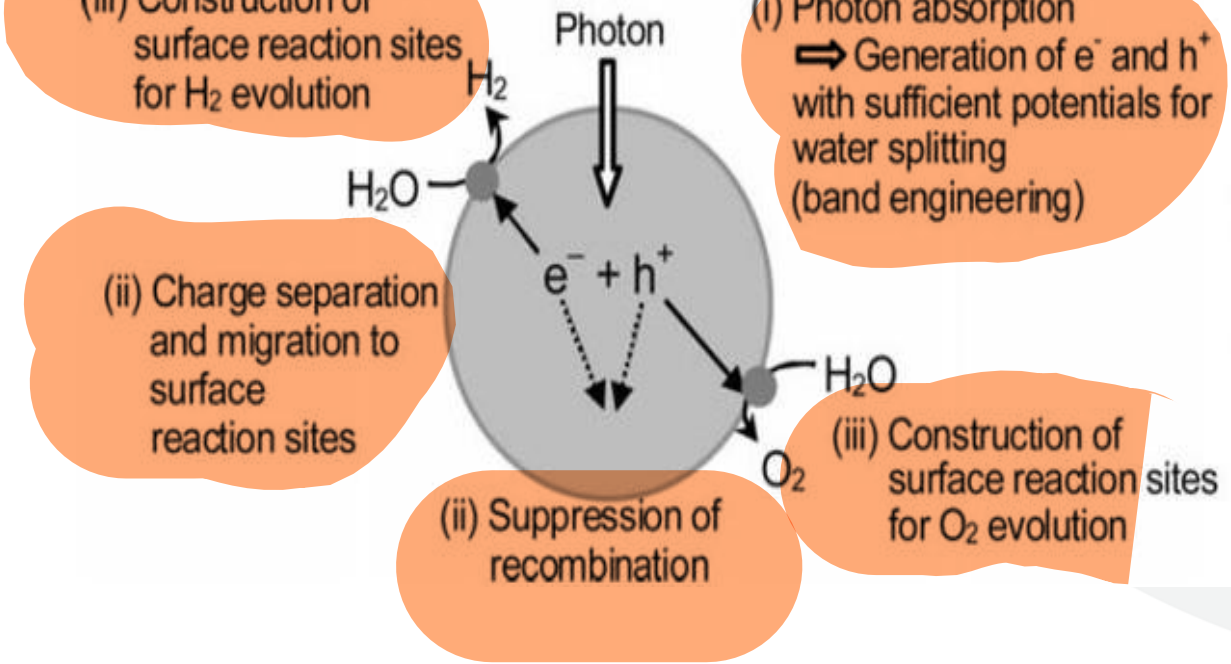
(iii) Construction of surface reaction sites for H_2 evolution

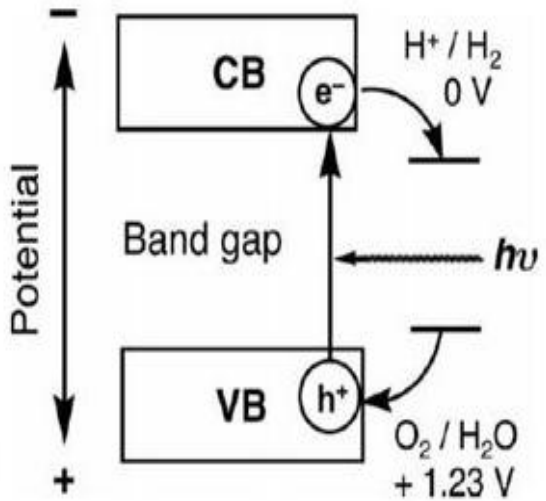
(i) Photon absorption ⇨ Generation of e^- and h^+ with sufficient potentials for water splitting (band engineering)

(ii) Charge separation and migration to surface reaction sites

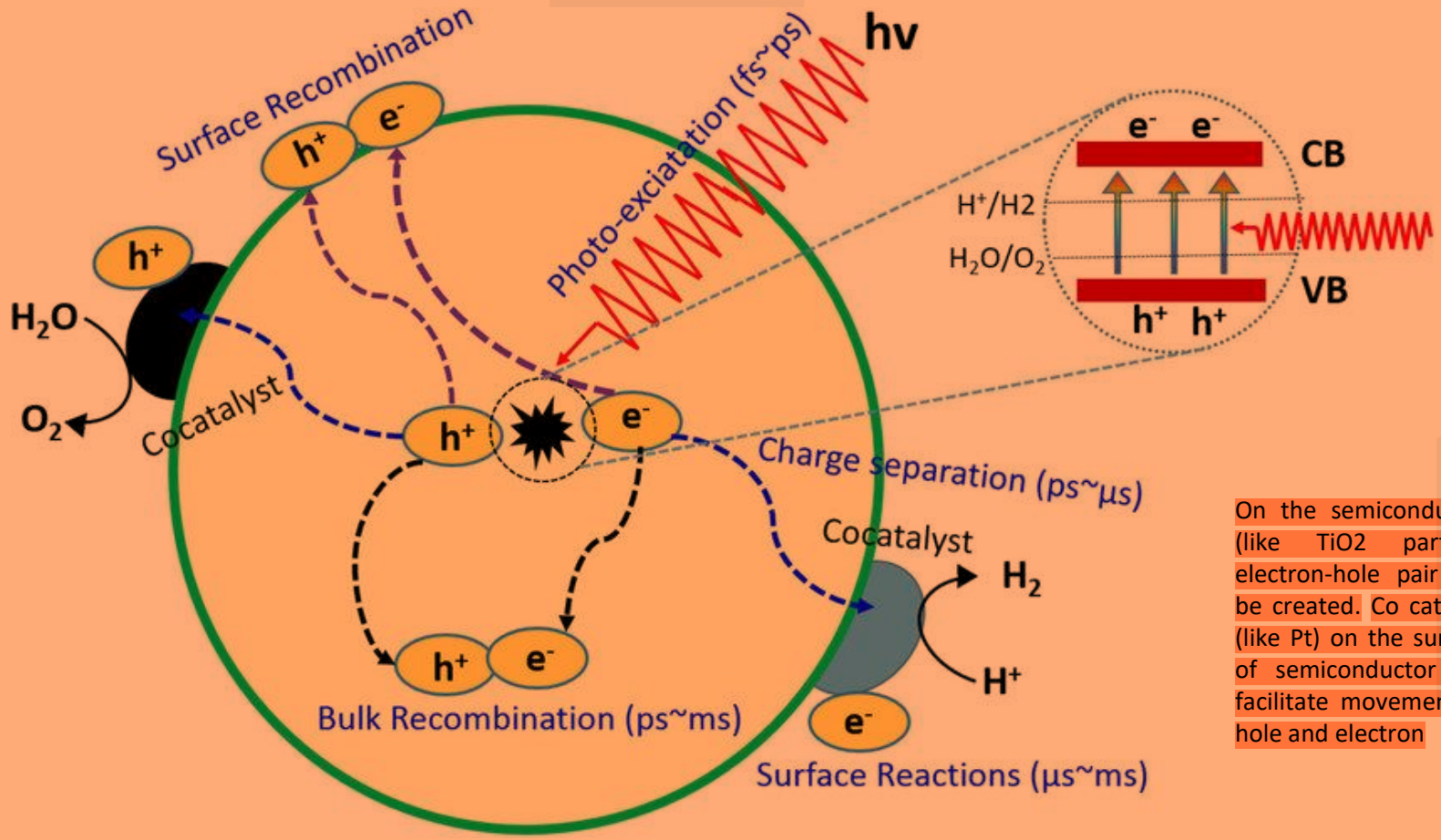
(ii) Suppression of recombination

(iii) Construction of surface reaction sites for O_2 evolution





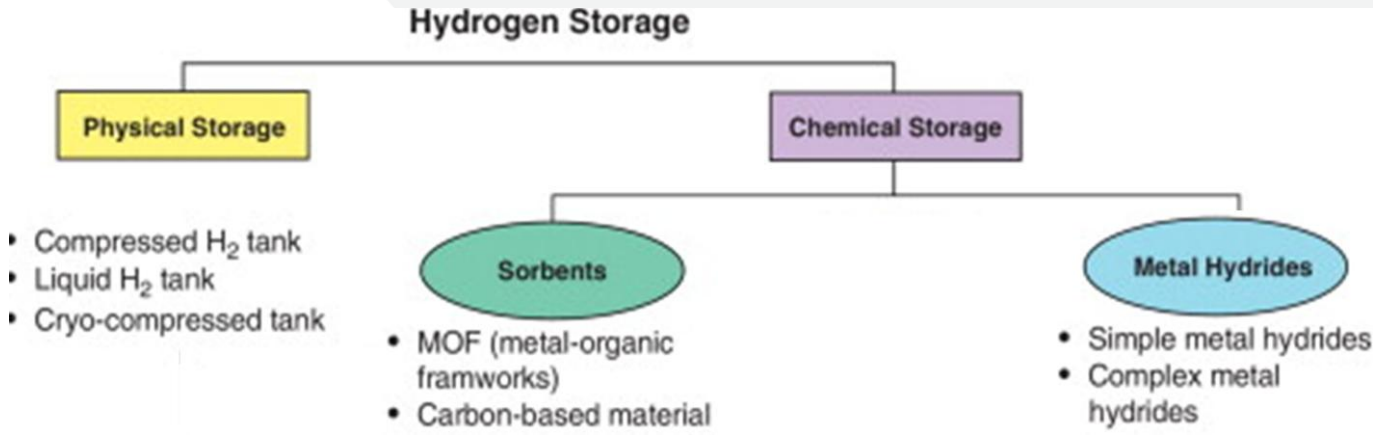
Mechanism of water splitting



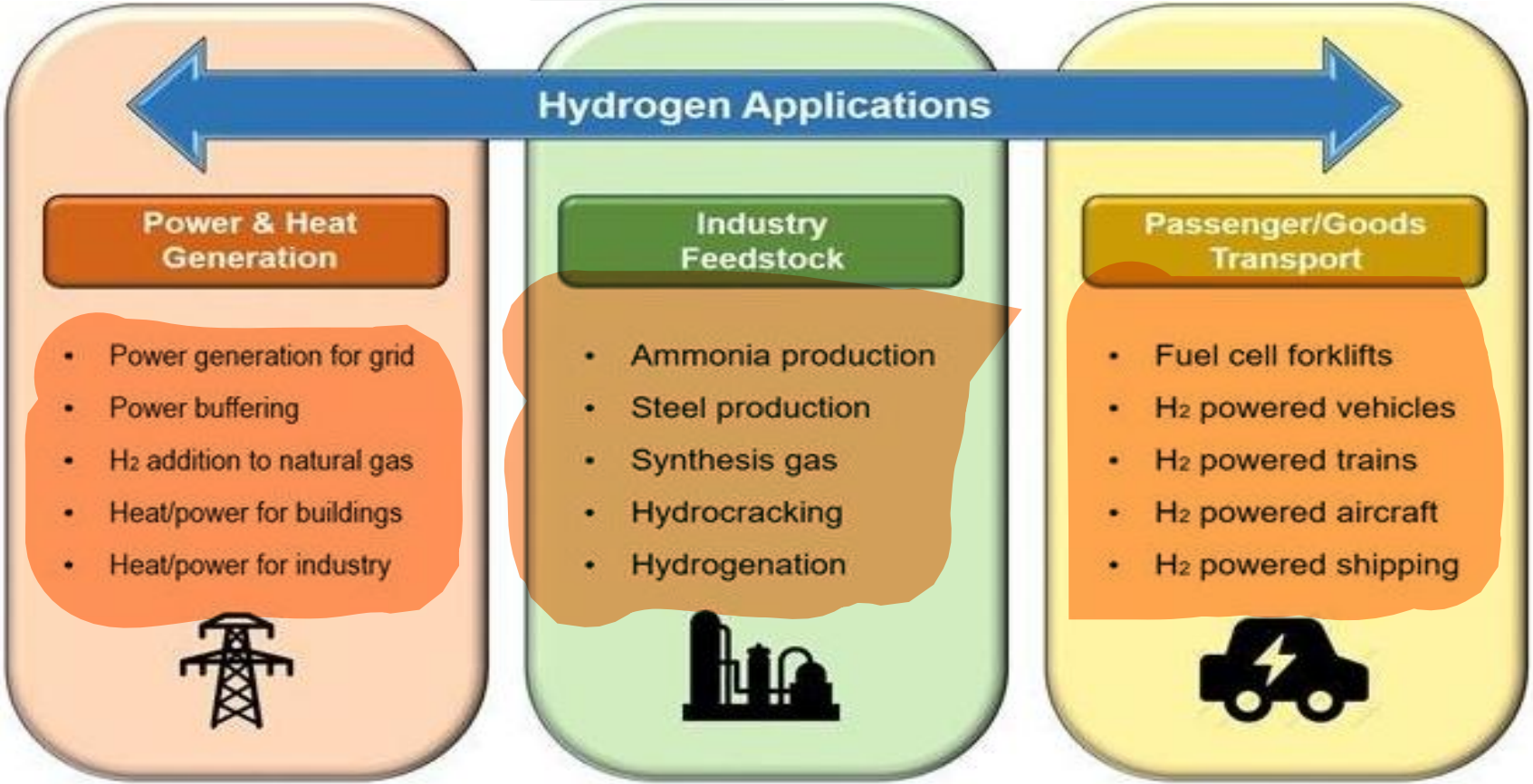
On the semiconductor (like TiO2 particle) electron-hole pair will be created. Co catalyst (like Pt) on the surface of semiconductor will facilitate movement of hole and electron

Hydrogen Storage

- Hydrogen can be stored either as a gas or as a liquid.
- Hydrogen gas storage typically requires the use of high-pressure tanks (350-700 bar or 5000-10,000 psi).
- liquid hydrogen storage requires cryogenic temperatures to prevent it boiling back into a gas (which occurs at −252.8°C).
- Hydrogen can also be stored on the surface of (adsorption) or within solid materials (absorption)



Applications of Hydrogen



Rocket Fuels

- Rocket fuels are nothing but Propellants- one which give the upward thrust.
- The homogeneous OR heterogeneous mixture of fuels and oxidizers used in rockets as a fuel, on burning produce high thermal energy and large quantity of hot gasses (propulsion mass) having very high exit velocities to give thrust or propulsion to rockets, are known as rocket propellants.
- Examples:

Oxidizers: Ammonium perchlorate (NH_4ClO_4), Ammonium nitrate (AN), Ammoniumdinitride (ADN). Metal per chlorates (MP)

Fuels: Hydrazine, Monomethyl hydrazine(MH), Unsymmetrical dimethyl hydrazine(UDMH), Kerosine, alcohol etc



Rocket Fuels

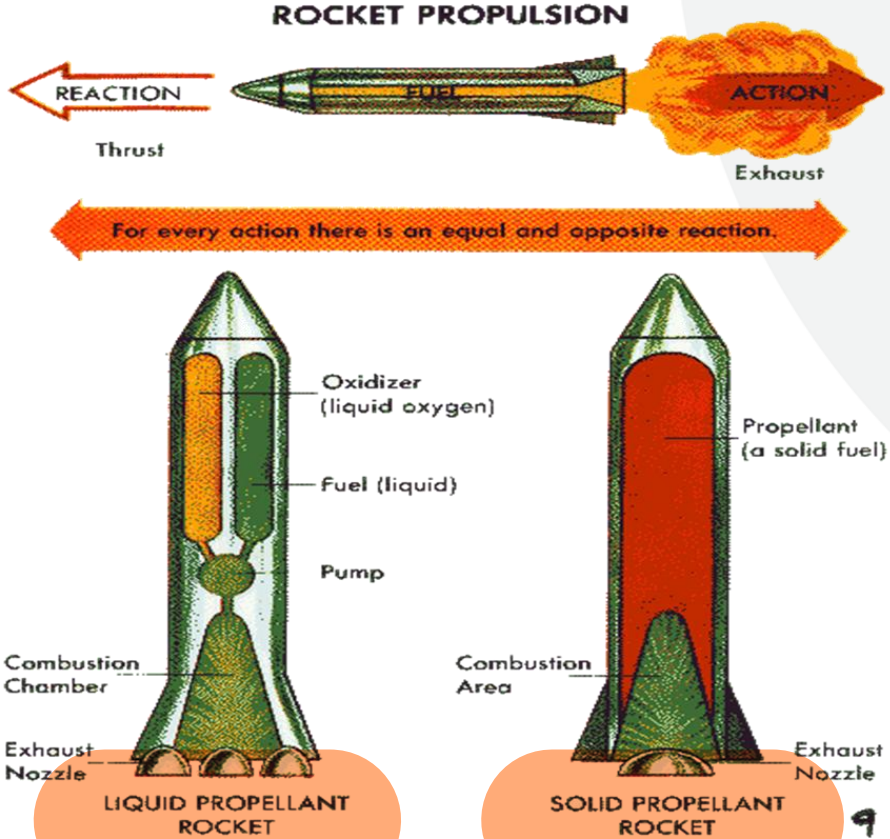
Characteristics of good Propellant

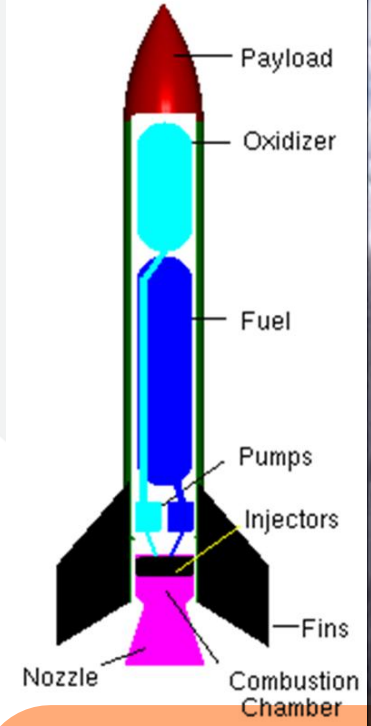
- It should produce high thermal energy on combustion.
- It should produce large amount of hot gasses (Propulsive mass) with very high exit velocity.
- It should burn at slow and steady rate.
- It should be stable to handle and store under ordinary conditions.
- It should be noncorrosive and non-hygroscopic.
- It should leave no solid residue after ignition



Principle of Rocket Propellants

- **Newton's third law of motion** : (Every action has an equal and opposite direction).



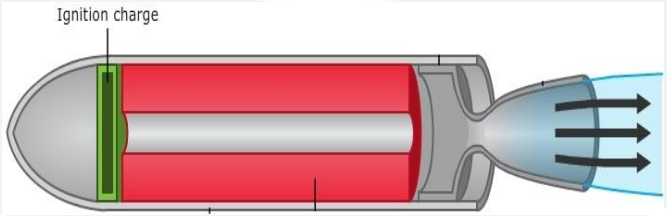




Classification of Rocket Propellants

Solid propellants

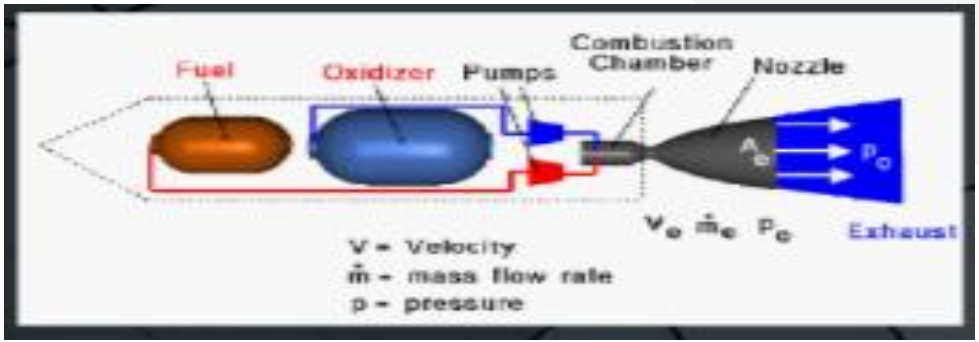
- Consist of a solid oxidizer and a solid fuel.
- Oxidizer and fuel are embedded in a polymer binder
- Solid oxidizers are ammonium perchlorate (AP) NH_4ClO_4, and aluminium powder, gun powder etc.
- The commonly used sold fuel is charcoal.
- Solid-propellants are much easier to store and handle than liquid-fuels which makes them ideal for military applications.



Classification of Rocket Propellants

Liquid propellants

- Consist of a liquid oxidizer and a liquid fuel.
- The fuel and oxidizers are filled in separate chambers and mixed in mixing chambers
- Liquid oxidizers are liquid oxygen, nitrogen tetroxide, hydrogen peroxide, Hydrazine and unsymmetrical dimethyl hydrazine.
- The commonly used liquid fuels are Hydrogen, Kerosene, Ethyl alcohol .



Classification of Rocket Propellants

Hybrid propellants

- Hybrid propellants are those propellants, which are a mixture of solid and liquid propellants.
- In these propellants, one of the two components (oxidizer and fuel) is solid (usually fuel) whereas the other is liquid (usually oxidizer).
- In a hybrid propellant rocket engine, the liquid part is injected into the solid part.
- The storage chamber of the solid part acts as the combustion chamber.

